

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458153.7006 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.015
Transit depth (Rp/Rs)^2	1.99 +/- 0.42 [%]
Orbital Inclination (inc)	83.47 +/- 0.51
Airmass coefficient 1 (a1)	0.993 +/- 0.011
Airmass coefficient 2 (a2)	0.0045 +/- 0.0055
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#1 - [572, 192]
Optimal Aperture	4.84
Optimal Annulus	13.9
Transit Duration (day)	0.0774 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.015 vs 0.116 (deviation 1.67σ)
Duration fit vs published	0.0774 d vs 0.09 d (deviation 1.3σ)
Mid-time O–C	-1.6 min
Depth SNR	9.4
Residual scatter	1.07 %

Light curve

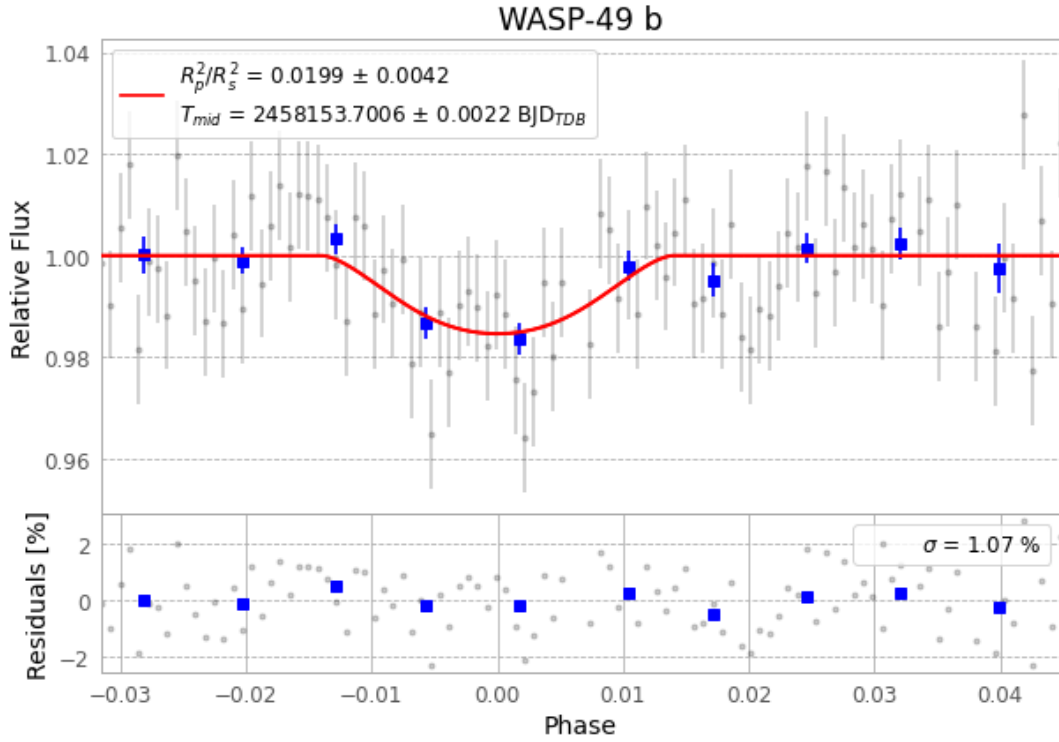


Figure 1 — FinalLightCurve_WASP-49 b_2018-02-03.png (SHA-256 4b21ed99cb44ec48...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [436, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 78ab6c0332274e72...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-49180204023612.FITS → WASP-49180204074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-180204.log (f5cffc91a26f...), FinalLightCurve_WASP-49 b_2018-02-03.png (4b21ed99cb44...), FinalLightCurve_WASP-49 b_2018-02-03.csv (6dfdf799bfd4...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 20:35Z.